

Merchant Link
FUNCTIONAL SPECIFICATION



BASE24-pos

Merchant Link Release 6.0

Version 1

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ACI Worldwide (Canada) Inc.
200 Wellington Street West
Suite 700
Toronto, Ontario M5V 3C7
(416) 813-3000

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Revision History

Version	Date	Author	Description
1.0	April, 2004	ME	Release 6.0 Updates

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1. INTRODUCTION

1.1. Objectives

The objective of this document is to describe the ACI Merchant Link product to give the reader an overall understanding of the product functionality. It is not intended to be a detailed description of the product software.

1.2. Background

ACI developed the BASE24-Merchant Link (ML) in 1992. The ML enables Canadian merchants to link up their BASE24-pos system to the Interac Debit network through a BASE24 Interac financial institution or third party processor or to link up to other Credit Networks. The ML has two components:

1. The ML Acquirer Interface, which runs on the merchant system.
2. The ML Acceptor Interface, which runs on the financial institution or third party processor system.

1.3. Business Requirements

Since 1991, the Interac Debit Network has allowed Financial Institutions (FI) to process debit transactions on behalf of merchants. Today there are Third Party Providers certified by Interac to provide this service as well.

Interac security regulations require the Financial Institutions to maintain control of the secure PIN Pads, regardless of their location or ownership, the merchant's debit terminals PIN Pads have to be controlled by the FIs debit systems. The merchant's terminals (debit or credit) can be controlled by either the merchant or the FI.

To fulfill this functionality, ACI developed the required FI merchant link, along with the required interface at the merchant system, as a standard ACI Canadian product known as Merchant Link.

1.4. Assumptions

Customers will be using ACI's Make to compile and segregate code, by using this tool the clients will be assured that their object code is up to date at all times.

1.5. Terminology

Acquirer:

The Acquirer ML interface refers to the interface implemented on the Merchant's computer system. The merchant is who ***acquires*** the transaction.

Acceptor:

The Acceptor ML interface refers to the interface implemented on the Financial Institution or Third Party Provider in their BASE24 system. The Acceptor BASE24-pos must include the switch to the Interac network. The FI or Third Party Provider is who ***accepts*** the transaction.

Issuer:

An Issuer, or card issuer, is an institution that ***issues*** cards or owns accounts that can be used in EFT transactions.

In resume the merchant is who acquires the transaction and it will send it to the Acceptor. The Acceptor can be a:

1. ***Financial Institution***

If the Acceptor is a financial institution, two scenarios can happen:

- a) The Acceptor is the Issuer of the card, and then the Acceptor will process the transaction.
- b) The Acceptor is not the Issuer of the card, and then transaction will be routed to the Issuer of the card.

2. ***Third Party Provider***

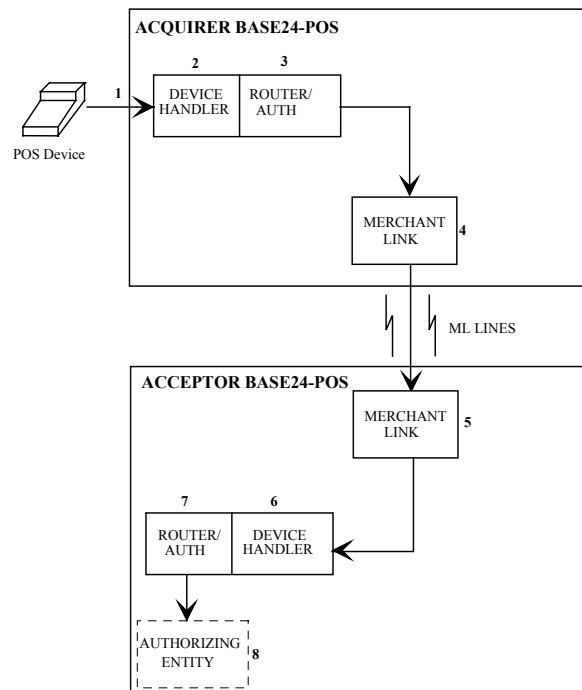
Third Parties will accept the transaction and route it to the Issuer of the card.

2. PRODUCT OVERVIEW

2.1. Overview of Product Architecture

The ML product consist of the following:

1. A standard release 6.0 SPDH/Router/Auth to handle ML transactions.
2. Two separate ML interface modules, one for the acceptor and one for the acquirer.
3. A Merchant Log Configuration File (MLCF), that keeps the static configuration of the Merchant Link.
4. A Merchant Totals File (MTF) that is used to accumulate totals for all transactions.
5. A Merchant Link Mapping File (MLMF) that is used to change the name of a terminal, so that two merchants can have the same terminal ID, but go to a different terminal on the acceptor.
6. A Store And Forward transaction file.



2.2. Acquirer Process

The Acquirer ML process acts as any other switch. Some of the main functions are:

- Logon to the acceptor ML process.
- Exchange keys with the ML acceptor process, if required.
- Accept a transaction from the SPDH. The message format is a PSTM, with two new tokens, CG and CI attached to it. Token CG, contains the message from the device, token CI contains reconciliation information.
- Place the transaction in suspense memory and add the transaction to the reconciliation totals.
- A message is formatted and sent from the acquirer to the acceptor. The exact format of the message can be chosen depends on the setting of the MLCF for that station.
- Encryption of confidential data (i.e. PIN and Account Balance).

2.3. Acceptor Process

The Acceptor ML process performs the following:

- Accepts the logon from the acquirer ML.
- Exchanges keys with the ML acquirer.
- Receives the message from the ML acquirer.
- Converts the message from the ML acquirer and converts it into an SPDH message.
- Puts the message into suspense memory, and starts a timer.
- Sends the message to an SPDH device handler. As far as the SPDH is concerned this is a normal POS transaction.

2.4. *Supported Features*

2.4.1. Supported Transactions

All the transactions from a POS device that the Acceptor Merchant Link allows are listed below:

- Normal purchase
- Pre-authorization purchase
- Pre-authorization purchase completion
- Merchandise return
- Balance inquiry
- Purchase with cash back
- Purchase adjustment
- Merchandise return adjustment
- Logon request
- Logoff request
- Close batch request
- Close shift request
- Close day request
- Employee subtotals request
- Batch subtotals request
- Shift subtotals request
- Day subtotals request
- Download request
- Handshake request
- Key change request

2.4.2. Store And Forward

If either the acceptor or the acquirer ML process is unable to deliver the transaction then it will be put in a store and forward file. The interface will attempt to deliver the transaction a predetermined number of times (which is set on the MLCF). If the transaction cannot be delivered after the predetermined number of tries it will be deleted from the SAF file, and an appropriate message sent to the logger.

2.4.3. ML MAC Support

If MAC generation/verification at the Merchant Link Interface is required for a given set of DPCs, those DPCs must be defined on BASE24 KEYF file as an interface record, with a process name of “ML”. A single DPC will share its MAC keys over multiple Merchant Link processes.

PIN block and field encryption, as well as POS device MACcing, are between the acceptor FI and the POS device and controlled by the acceptor's POS Terminal Definition File (PTDF).

The Merchant Link supports MACcing between the Merchant Link interfaces. This MACcing ensures that data other than the POS device transactions such as settlement totals and network management messages are not corrupted. The BASE24 ML interfaces both acquirer and acceptor, require hardware security modules to perform MACcing.

2.4.4. Concurrent Transaction Processing

The ML interface will support concurrent processing for credit transactions:

- The employee ID field in the SPDH header is unique, thus only one employee can have a transaction outstanding at any one time.
- There can be no MAC and no PIN sent, this means that an Interac transaction cannot be sent; only credit transactions can be sent.
- When a response is received, processing flag 3 on the SPDH header will be set to 1.
- Read-only PTDF records.

2.4.5. NCS Commands

The ML Acceptor and Acquirer handle the following NCS Commands:

LOGON <dpc num>	Send a logon network management message to the DPC in question.
LOGOFF <dpc num>	Send a logoff network management message to the DPC in question.
NEWMACKEY <dpc num>	Generate new MAC working keys and send them to the DPC in question.
PERFORMANCE <dpc num>	Log performance statistics for the last five performance intervals (defined in the MLCF) to the BASE24 log.
STATUS <dpc num>	Logs file and message throughput status for a given DPC. If <dpc num> is not included, all DPCs supported by the ML will be displayed.

**TOTALS (<date> <settle
entity> <card type>)**

Based on the MTF totals for the entered acceptor BASE24 posting date, send a reconciliation totals request for the entered settlement entity (ML type 0800, code 201 or ISO 0500 message)

9501/WARMBOOT

Reinitializes the ML process by closing and reopening all files and rebuilding all tables.

2.4.6. Pre-Authorization Support

The Merchant Link product supports pre-authorization request and pre-authorization completion transactions.

Pre-authorization requests is accepted and forwarded to the proper issuer by the Acceptor. If the Acceptor is unable to forward the request, then the transaction will be declined or time out.

When a merchant sends a pre-authorization completion, the ML Acquirer acknowledges the completion to the POS device, stores the completion in the SAF file (only if link is down) and forwards the completion to the Acceptor FI.

In order to handle a pre-authorization completion on the Acquirer ML, the Store and Forward (SAF) file is used to store any pre-authorization completion that cannot be sent to the Acceptor real-time. The SAF file contains pre-authorization completion and "must complete" reversals.

2.4.7. Acquirer / Acceptor Token support

Starting in release 5.0 the ML Acquirer and ML Acceptor modules support Message Tokenization.

There are two tokens defined:

- Token CG POS Device Message Token
- Token CI POS Merchant Data Token

When the ML processes an internal transaction or administrative reply, the suspense record must first be located. Appended to the PSTM, a token named "CG" will carry data enabling retrieval of the suspense record in a 6.0 network.

The ML must know the reconciliation entity when calculating financial reconciliation totals. Appended to the PSTM, a token named "CI" will contain the reconciliation entity (RCNCL-ENT).

2.5.1. ML Format

The ML format consists of a message header, common to all messages, that contains information about the Acquirer and about the type of processing required for this transaction. This header is followed by one of two message structures: either a network management message or an Acquirer POS device Message.

Network management messages are sent only between requesting and responding ML processes. These messages can be logons, logoffs, echo tests, MAC key exchanges or reconciliation messages.

Financial Transactions Structure

ML Header	SPDH Header	FIDs
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Network Management Transactions Structure

ML Header	Network Management Fields
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2.5.1.1. Merchant Link Header

The Merchant Link header in the ML format consists of 11 fields, totaling 75 bytes. It is not configurable. The fields and their respective lengths are listed below:

Field Name	Len	Type	Description
IDENTIFIER	2	a	Identifies message as a ML message. This is required for the Device Handler bound module. It's value is always "ML"
DPC NUMBER	4	n	DPC number originating this transaction.
MESSAGE TYPE	4	n	The type of message sent, having one of the following values: 0000 - POS device administrative transaction 0200 - On-line transaction request 0210 - On-line transaction reply 0220 - Forced post transaction request 0230 - Forced post transaction reply 0430 - Reversal reply 0800 - Network management request 0810 - Network management reply
TRACE NUMBER	6	n	An internal trace number, set by the transaction acquirer, sent back unchanged on replies to be used for message tracking and the detection of duplicate messages

DATE	6	n	Local date, in the form YYMMDD, at which the message was sent on from the ML.
TIME	8	n	Local time, in the form HHMMSSSS, at which the message was sent to or from the ML.
RECONCILIATION ENTITY	1	n	One digit value, assigned to a given PIN Pad by PTDF indicating the settlement entity for which reconciliation totals are being accumulated or exchanged. Possible values are: 0 - Totals not accumulated for this transaction (initial value on requests) 1 - Totals accumulated by DPC 2 - Totals accumulated by Retailer ID 3 - Totals accumulated by Retailer Group 4 - Totals accumulated by Terminal ID 5 - Totals accumulated by Clerk ID
SETTLEMENT IDENTIFIER	25	n	Entity for which settlement totals are being kept. (see Reconciliation Entity, above). Blank filled on requests and NMM messages other than reconciliation. On requests and reconciliation messages, it may have one of the following values: blanks - Settlement is for entire DPC (or no settlement done) Retailer ID - Settlement is for one retailer within the DPC. Retailer Group - Settlement is for one retailer group with the DPC. Terminal ID - Settlement is for one local DPC terminal ID. Clerk ID - Settlement is for one employee id under one retailer in the DPC.
CARD TYPE	2	an	Two character mnemonic identifying the card type (e.g. “V” = VISA, “AX” = American Express, etc.) to which this transaction applies. On a reconciliation transaction (NMM code 201), this field may be blanks to indicate that the totals sent are for all card types.
MAC INDEX	1	n	Identifies which of the two MAC working keys will be used to verify this message. This field is optional.
MAC	16	h	Message Authentication Code, built from the remainder of the header fields, to be used to verify this message between the ML and the acquirer Host. It should not be confused with the MAC on the terminal transaction request. This field is optional

In order to use this format the field “EXT MESSAGE FORMAT” should be set to “0”.

For more information on the SPDH format, consult the “BASE24 Standard Product Device Handler” Manual.

2.5.2. ML ISO message Format

Based on the message format indicator on the MLCF (Merchant Link Configuration File), individual DPCs may be configured to use ISO 8583 standard messages. This involves a functional mapping between the fields in MLSPDH format and the ISO format.

In order to support the ISO message format the ML interface will translate messages from the ISO Message format to the MLSPDH format. The details can be found in "Appendix A - ISO to MLSPDH Message Mapping".

When this method of communication is used then the MLCF field "EXT MESSAGE FORMAT" should be set to "2" if ISO 8583 is to be used in the text mode and "3" if the ISO 8583 binary mode is to be used.

2.5.3. BMO (Bank of Montreal) ISO Format

This is a variant of the ISO 8583 standard is used when communicating with the Bank of Montreal. The details of this message format can be seen in "Appendix B - Bank of Montreal ISO to MLSPDH Mapping".

When this method of communication is used then the MLCF field "EXT MESSAGE FORMAT" should be set to "4" if ISO 8583 is to be used in the text mode and "5" if the ISO 8583 binary mode is to be used.

2.6. Merchant Totals File (MTF)

The MTF (Merchant Totals File) screen displays the totals accumulated for a given Settlement Entity on a given day.

The following reconciliation entities are used in the ML interface. They are defined on the acquirer's PTDF record and when the transaction is approved the amounts are added to the MTF record. The following reconciliation entities are defined in the ML interface:

- 1 - Totals accumulated by DPC
- 2 - Totals accumulated by Retailer ID
- 3 - Totals accumulated by Retailer Group
- 4 - Totals accumulated by Terminal ID
- 5 - Totals accumulated by Clerk ID

There are two methods to display totals:

- Look at the reconciliation totals online using the MTF Requester/Server.
- NCS Totals Command

2.7. Merchant Link Mapping File (MLMF)

The MLMF file allows the acceptor FI to map the merchant (or acquirer) PIN Pad identifier (FID t) to a Terminal ID defined on the acceptor's PTDF. This is useful for the transaction acceptor, since it allows them to change terminal Id's without impacting the acquirer. This allows the term-id on the request to be mapped to a new value defined in the MLMF.

This information is entered using the MLMF requester/server. For more information please consult the "ML Release 6.0 Installation Guide".

2.8. Merchant Link Configuration File (MLCF)

The MLCF resembles the Host or Interchange Configuration Files (HCF/ICF). MLCF contains control information for each DPC connection supported by a given ML process.

The MLCF is broken down into three sections:

- Timers (Network Management, performance).
- Processing flags.
- Configuration of stations per DPC.

2.9. Base24 Module Merchant Link Support

In order to support Merchant Link some BASE24 modules contain Merchant Link specific functionality. A brief description of this functionality follows, for further information see the relevant product manual.

2.9.1. Device Handler Module (SPDH) – Common functionality

The SPDH module supports the following ML functionality:

- If a transaction is received that includes FID t (PIN Pad ID), the PTD will be read using this ID, rather than the terminal ID in the SPDH header.
- Settlement totals control fields, such as retailer id, are present on the PSTM, but not on the device reply. For this reason, transaction replies sent to the ML will be in the same format as requests (i.e. PSTM followed by SPDH message).
- FID u optionally is sent containing the BASE24 posting date in the form, YYMMDD.
- In order to distinguish between acceptor and acquirer processing, a LCONF parameter POS-PASSTHRU-PROCESS-TYPE will be set to either 'ACQ' (acquirer processing) or 'ACC' (acceptor processing).

2.9.2. Device Handler Module (SPDH) - Acquirer Functionality

The SPDH supports the following functionality for support of an acquirer ML interface:

- New transaction types, “M” (ML financial) and “L” (ML administrative) are accepted and processed as their non-ML counterparts, “F” (financial) and “A” (administrative).
- MAC verification will not be done for these transactions.
- The PSTM field, DEST, initializes to the value, “ML “, to identify the transaction as destined for an ML interface.
- Forced posts, such as pre-auth completion, is a special case, since the DH normally just acknowledges their receipt and forwards the message to the authorizer.

The SPDH receives a pre-authorization completion (220) from a POS terminal. The Device Handler then sends an acknowledgment back to the terminal with the SPDH response (Resp Code 000) to complete the transaction. The device handler acknowledges all Pre-Authorization completions.

The Router/Auth logs the completion (220) to the PTLF file as an approved transaction and it routes the completion to the Acquirer ML Interface.

- The ML interface sends a transaction reply appended as a token to the PSTM. This reply, rather than one built from the PSTM and PTD suspended fields, is sent to the terminal.

2.9.3. PTLF Extract Module (EXT)

The PTLF extract module supports functionality to process ML reconciliation records.

2.10. POS Terminal Data (PTD)

The PTD file contains 2 fields specific to ML processing.

- Destination DPC
- Reconciliation Entity

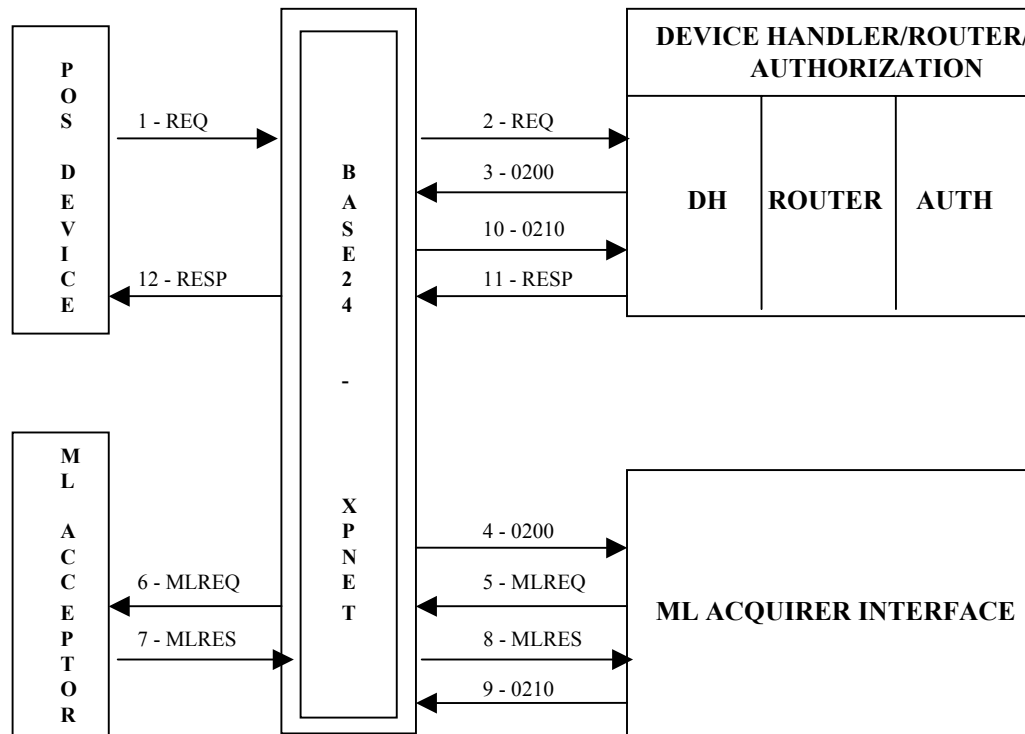
More information can be found on these fields in the BASE24-pos Files Maintenance Manual.

3. MESSAGE FLOWS

This section provides examples of transaction message flows through the BASE24 Merchant Link system under various types of processing situations. The flows assume a BASE24 ML Acceptor and Acquirer are present.

3.1. *Approved/Denied Financial Transaction*

In the diagram below, the transaction is a ML financial transaction request, the ML interface allows a merchant DPC to acquire transactions from Standard POS Devices that they themselves cannot authorize, such as Interac Debit traffic. The following is an example of a ML financial transaction:



Steps

Processing

1-2

The POS device sends the SPDH transaction request to BASE24 ML Acquirer as a native-mode message, where it is received by the BASE24 XPNET and forwarded to the Device Handler with a message type of “M”, or Merchant Link Financial transaction.

- 2-3 The Device Handler module processes the request as normal, ignoring the PIN Pad identifier (FID t) and formats the transaction request into a POS Standard Internal Message (PSTM), with the DEST field set to “ML” to identify the transaction as belonging to an external FI, adds the two ML tokens, sets a response timer, and passes the transaction request as a 0200 message to the Router/Auth module.

The Router/Auth module also processes the request as normal, retrieving the Retailer and card specific information and finding the ML BASE24 symbolic process name within the Auth Selection Table (AST) or Institution Definition File (IDF). The PSTM request is at this point fully formatted for authorization, and is also appended with the original device request (token CG).

- 4-5 The acquirer ML process receives the PSTM, retrieves the CG token, original request (not including the PSTM) to the acceptor ML, prefixed by a message header. This request may be in ISO format.

- 6-7 Upon receiving the transaction request, the acceptor ML removes the ML header and sets it's own inbound timer, suspends the transaction. If a PIN Pad mapping is required for this DPC, FID t (PIN Pad ID) is extracted from the request and mapped to an unique internal ID. The request, in the format of a normal transaction request, is sent to the DH as if the ML is a POS device.

The Acceptor Device Handler, detecting the presence of FID t, uses the PIN Pad ID to retrieve a POS Terminal Data (PTD) record and processes the transaction in the current manner.

The Acceptor Router/Auth module also treats the incoming transaction as one from the FI's own terminal.

The request is authorized by any Host, Interac, or Switch currently supported in the FI's BASE24-pos environment.

The Host, Interac or the Switch approves the transaction request, formats a 0210 response message, and sends the message to the Acceptor Interface.

The Acceptor ML process retrieves the suspended transaction, deletes the inbound timer, creates a ML response based on the 0210 response message and sends it to the ML acquirer.

The Totals in the MTF file will be updated.

- 8-9 The ML acquirer interface process retrieves the suspended record, reformats the response message into a PSTM 0210 response and CG token message, deletes the outbound timer, and sends the response message to the Device Handler/Router/Authorization process.

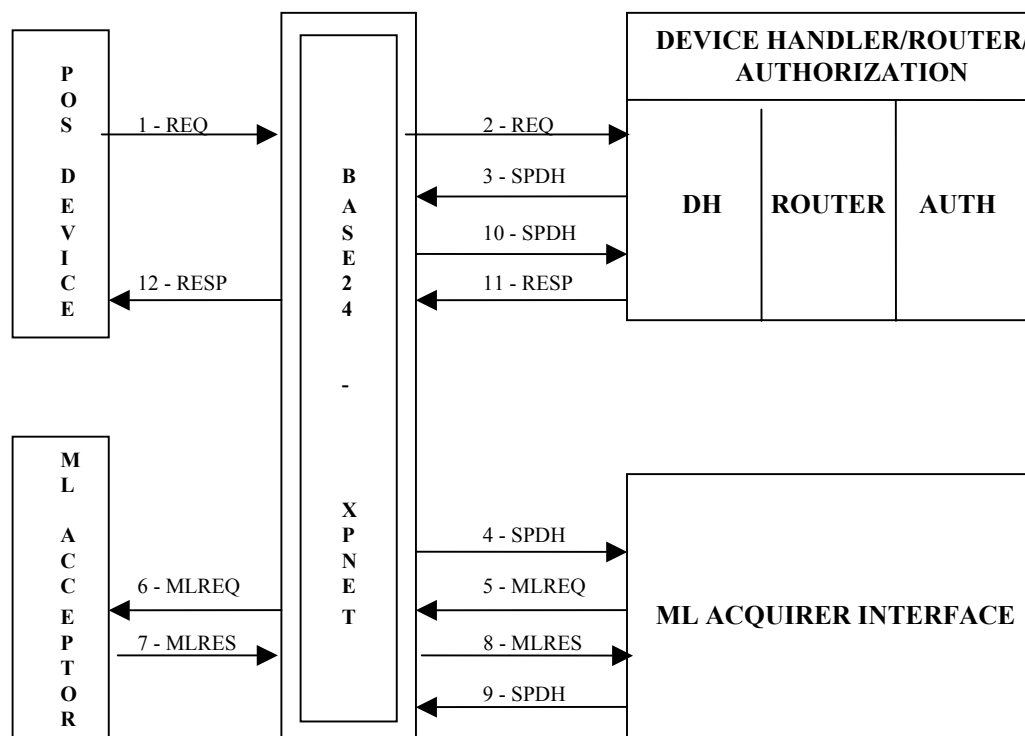
The Totals in the MTF file will be updated.

- 10 The Authorization module logs the transaction to the POS Transaction Log File (PTLF) including the CI token.

- 11 -12 The Device Handler module reformats the 0210 response and CG token message into a SPDH native-mode format, deletes the response timer, and sends the response to the POS device.

3.2. Terminal Administrative Transaction

In the diagram below, the transaction is a ML administrative transaction.



Steps

Processing

- 1-2 The POS device sends the transaction request to BASE24 ML Acquirer as a native-mode message, where it is received by the BASE24 XPNET and forwarded to the Device Handler with a message type of “L”, or Merchant Link Administrative transaction.
- 2-3 The Device Handler sends the Admin request to the ML Acquirer Interface as a SPDH message. Note that the Administrative transaction does not use the PSTM internal message like the Financial.
- 4-5 The acquirer ML process sends the SPDH message request to the ML Acceptor.
- 6-7 Upon receiving the administrative transaction request, the acceptor ML responds according with the type of administrative transaction to the ML Acquirer.

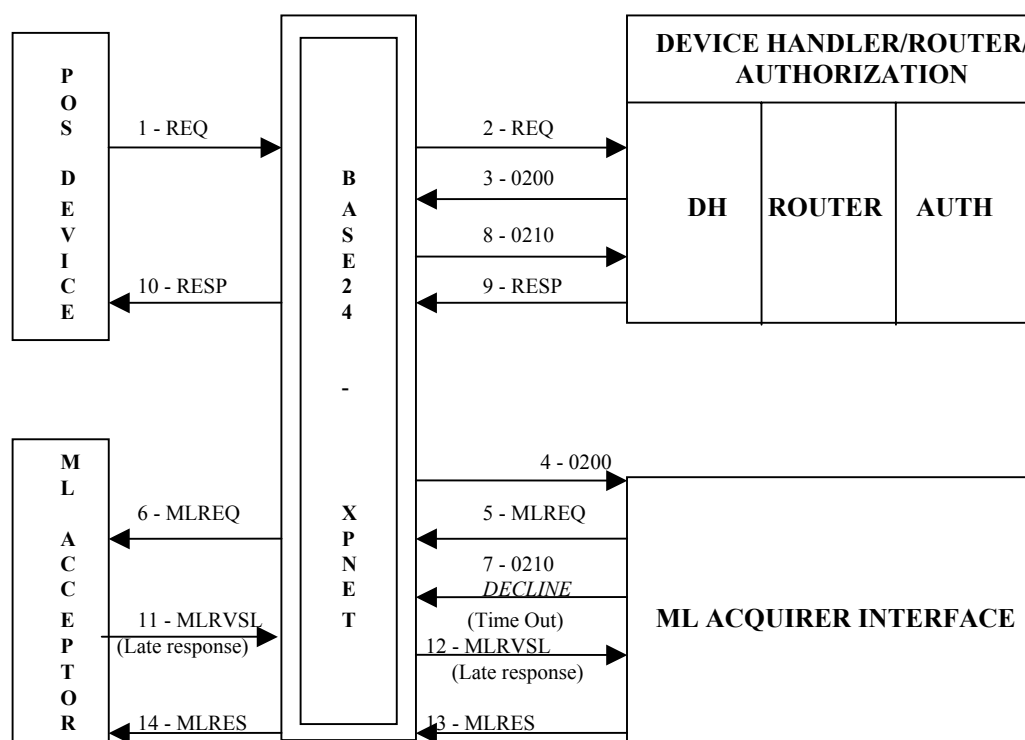
- 8-9 The ML acquirer interface process sends the administrative response back to the SPDH

- 10-11 The Device Handler module responds to the terminal with the response message that was generated by the Acceptor ML.

- 12 The terminal gets the response to the administrative message.

3.3. Reversal on Late Response from the Acceptor

In the diagram below, the transaction is a ML financial transaction request, the ML Acceptor is the primary authorizer. Although the ML Acceptor approves the transaction, the response is late and the transaction times out. When the transaction times out, the ML Acquirer Interface process automatically denies the transaction. Therefore, when BASE24 receives the late approval from the Acceptor, The Acquirer sends a reversal to the Acceptor because the transaction was already denied.



Steps

Processing

- 1-2 The POS device sends the SPDH transaction request to BASE24 ML Acquirer as a native-mode message, where it is received by the BASE24 XPNET and forwarded to the Device Handler with a message type of “M”, or Merchant Link Financial transaction.
- 2-3 The Device Handler module processes the request as normal, ignoring the PIN Pad identifier (FID t) and formats the transaction request into a POS Standard Internal Message (PSTM), with the DEST field set to “ML” to identify the transaction as belonging to an external FI, adds the two ML tokens, sets a response timer, and passes the transaction request as a 0200 message to the Router/Auth module.

The Router/Auth module also processes the request as normal, retrieving the Retailer and card specific information and finding the ML BASE24 symbolic process name within the Auth Selection Table (AST) or Institution Definition File (IDF). The PSTM request is at this point fully formatted for authorization, and is also appended with the original device request (token CG).

- 4-5 The acquirer ML process receives the PSTM, retrieves the CG token, creates a ML request, suspends the transaction, sets an outbound timer, and sends the original request (not including the PSTM) to the acceptor ML, prefixed by a message header. This request may be in ISO format.
- 6 Upon receiving the transaction request, the acceptor ML removes the ML header and sets it's own inbound timer, suspends the transaction. If a PIN Pad mapping is required for this DPC, FID t (PIN Pad ID) is extracted from the request and mapped to a unique internal ID. The request, in the format of a normal transaction request, is sent to the DH as if the ML is a POS device.
- The Acceptor Device Handler, detecting the presence of FID t, uses the PIN Pad ID to retrieve a POS Terminal Data (PTD) record and processes the transaction in the current manner.
- The Acceptor Router/Auth module also treats the incoming transaction as one from the FI's own terminal.
- The request is authorized by any Host, Interac or Switch currently supported in the FI's BASE24-pos environment.
- 7 The outbound request timer expires before the ML acceptor returns a response. Therefore, the ML Acquirer Interface process declining the transaction request. The Acquirer Interface process generates a 0210 response message in internal format and sends the 0210 response message to the Device Handler/Router/Authorization process. At the same time step 11 is initiated.
- 8-9 The Device Handler/Router/Authorization process logs the transaction to the POS Transaction Log File (PTLF). The Device Handler reformats the 0210 response message into native mode format, deletes the response timer and sends the response message to the POS device.
- 11-12 This step happens concurrently with step 7. Since the transaction request was already declined, the Acquirer Interface process generates a SPDH Controller "C" reversal with the information received on the transaction response and sends the reversal to the Acceptor. The reversal allows the authorizer to update its database appropriately.

13-14 The acceptor sends the reversal. The ML acquirer does not log it as it has already logged the transaction as a controller failure. The ML Acceptor has logged it as 420, the terminal has been told that the transaction has failed due to an error. The three parts are balanced.

Note: Even if the last two steps occur in the reverse order then the totals will still be correct.

3.4. *Terminal Reversal Transaction*

Reversals Request from the Terminal can be as follows:

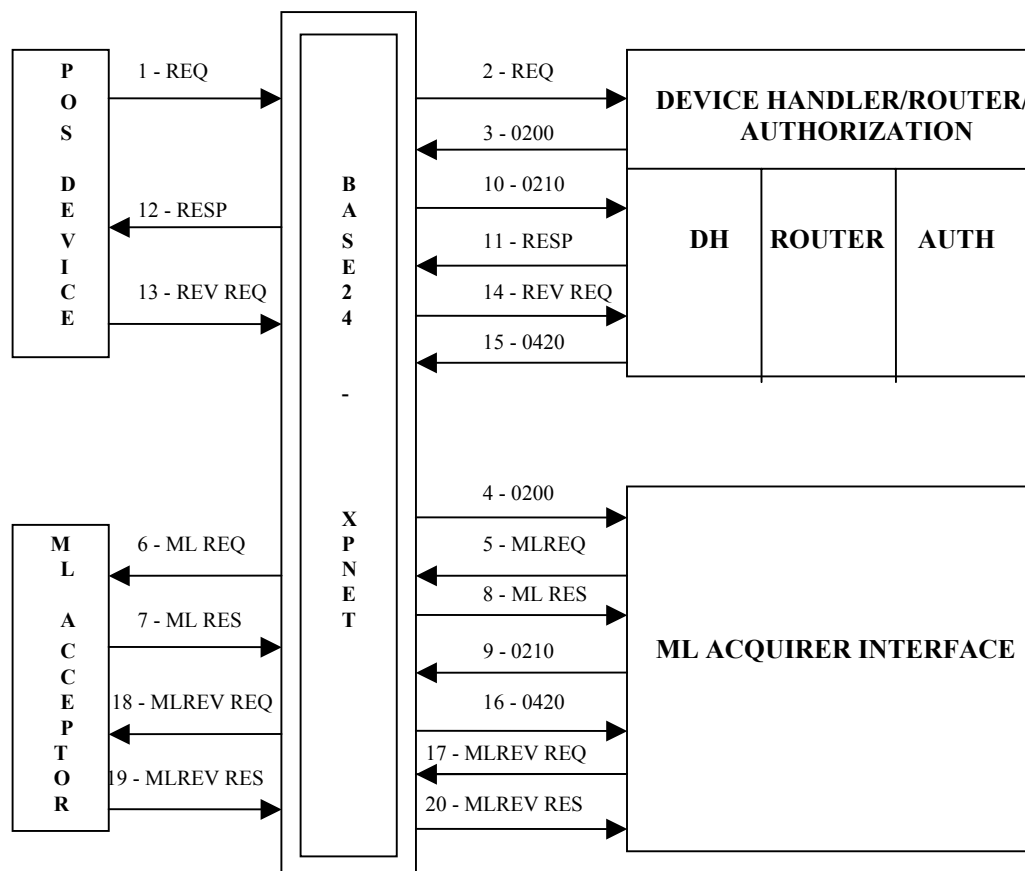
- The **MAC reversal (R)** is produced when the terminal receives a response message and the MAC value calculated by the POS/PIN pad device is not the same as the one in the response message.
The response message is then sent back to the SPDH Device Handler as received, with the exception of the field SPDH Header Message Subtype equal to “R”, thus reversing the transaction.
- The **Terminal or Controller reversal (C)** is produced when the POS device or the POS controller had a failure and the transaction can not be completed and the transaction must be reversed.

The response message is then sent back to the SPDH as received, with the exception of the field SPDH Header Message Subtype equal to “C”, thus reversing the transaction.
A controller might place a C in this field if the controller is unable to send the response to the individual terminal that generated the transaction. A terminal might place a C in this field if the terminal has already timed out before receiving the BASE24 response.

- The **Terminal Decryption reversal (D)** is produced when a balance inquiry transaction must be reversed due to a data decryption error by the POS/PIN pad device.
The response message is then sent back to the SPDH as received, with the exception of the field SPDH Header Message Subtype equal to “D”, thus reversing the transaction. This will occur ONLY with balance inquiry transactions when the available balance cannot be decrypted.

The following is an example of a ML financial transaction with a Reversal Request from the POS Device, steps from 1 to 12 are identical to the Approved/Denied flow described in section 3.1.

Terminal Reversal Transaction (Cont.)



Steps

Processing

13-14

The POS device sends the SPDH Reversal transaction request to BASE24 ML Acquirer as a native-mode message, where it is received by the BASE24 XPNET and forwarded to the Device Handler with a message type of “M” or Merchant Link Financial transaction and message subtype “R” (MAC) or “C” (Controller) or “D” (Decryption).

14-15

The Device Handler module receives the reversal request and formats the reversal transaction request into a POS Standard Internal Message (PSTM), with the DEST field set to “ML” to identify the transaction as belonging to an external FI and passes the reversal transaction request as a 0420 message to the Router/Auth module.

The Router/Auth module logs the reversal transaction routes it to the Merchant Link Interface based on the routing parameters. The PSTM reversal request is also appended with the original device reversal request (token CG).

- 16-17 The acquirer ML process receives the reversal PSTM (0420), retrieves the CG token, creates a ML request and checks the connection with the Acceptor that needs to be UP and logically Logon ON.
If the above is true, it suspends the reversal transaction, sets an outbound timer and sends the reversal request (0420) (not including the PSTM) to the acceptor ML, prefixed by a message header in either SPDH or ISO format.
If the ML interface does not have communication or is logically Logoff or the reversal request sent to the Acceptor times out, it will be added to the SAF file for later transmission.

Note: The ML interface has a SAF cycle that is being driven by the MLCF file SAF timer and SAF retries parameter.

- 18-19 Upon receiving the reversal transaction request (0420), the Acceptor ML formats a transaction response (0430) and sends it to the ML Acquirer.

The ML Acceptor also removes the ML header and forward the SPDH reversal transaction request to the Device Handler/Router/Auth as if the ML is a POS device.
If a PIN Pad mapping is required for this DPC, FID t (PIN Pad ID) is extracted from the request and mapped to a unique internal ID.

The Acceptor Device Handler, detecting the presence of FID t, uses the PIN Pad ID to retrieve a POS Terminal Data (PTD) record and processes the reversal transaction.

The reversal request is authorized by any Host, Interac or Switch currently supported in the FI's BASE24-pos environment.

The Totals in the MTF file will be updated.

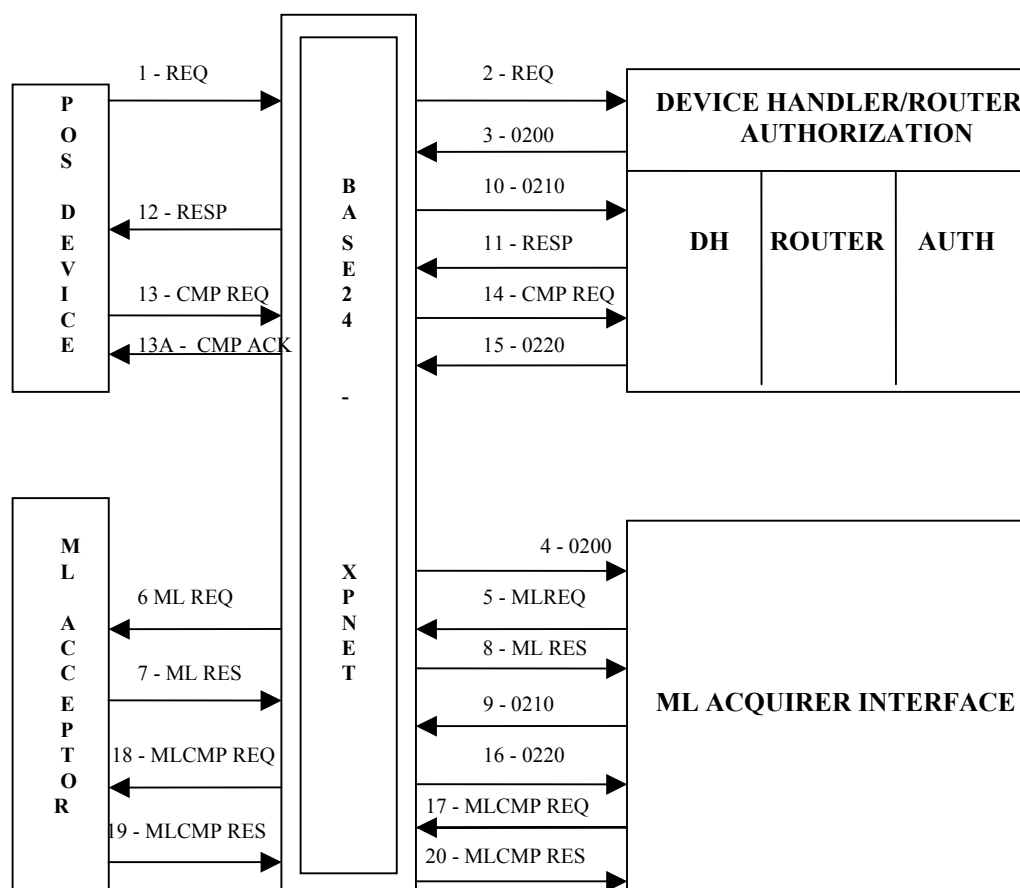
- 20 The ML Acquirer receives a transaction response (0430) for a real time or SAF cycle. Then, it deletes the outbound timer or the SAF record will be deleted.

The Totals in the MTF file will be updated.

If the outbound request timer expires before the ML acceptor returns a reversal response (0430), a SAF timer will be set for a SAF cycle and a SAF record will be added if the first attempt was real time.

3.5. Pre-Authorization, Pre-Authorization Completion

In the diagram below, the transaction is a ML financial transaction request is identical to the Approved/Denied flow described on Section 5.1 from steps 1 to 4.



Steps

Processing

- 4-5 The acquirer ML process receives the PSTM, retrieves the CG token, creates a ML request, suspends the transaction, sets an outbound timer and sends the original request as a 0100 message since it is a pre-auth transaction (not including the PSTM) to the acceptor ML. The request is prefixed by a message header and may be in ISO format.
- 6-7 Upon receiving the transaction request 0100, the acceptor ML removes the ML header and sets it's own inbound timer, suspends the transaction. If a PIN Pad mapping is required for this DPC, FID t (PIN Pad ID) is extracted from the request and mapped to a unique internal ID. The

request, in the format of a normal transaction request, is sent to the DH as if the ML is a POS device.

The Acceptor Device Handler, detecting the presence of FID t, uses the PIN Pad ID to retrieve a POS Terminal Data (PTD) record and processes the transaction in the current manner.

The Acceptor Router/Auth module also treats the incoming transaction as one from the FI's own terminal.

The request is authorized by any Host, Interac or Switch currently supported in the FI's BASE24-pos environment.

The Host, Interac or the Switch approves the transaction request, the UAF/CAF file is update with the pre-authorization hold information. Then, it formats a 0210 response message, and sends the message to the Acceptor Interface.

The Acceptor ML process retrieves the suspended transaction, deletes the inbound timer, creates a ML response as a 0110 since it is a pre-authorization transaction based on the 0210 response message and sends it to the ML acquirer.

The Totals in the MTF file will be updated.

- 8-9 The ML acquirer interface process retrieves the suspended record, reformats the response message into a PSTM 0210 response and CG token message, deletes the outbound timer, and sends the response message to the Device Handler/Router/Authorization process.
- The Totals in the MTF file will be updated.
- 10 The Authorization module logs the transaction to the POS Transaction Log File (PTLF) including the CI token.
- 11 -12 The Device Handler module reformats the 0210 response and CG token message into a SPDH native-mode format, deletes the response timer, and sends the response to the POS device.
- 13-14 The POS device sends the SPDH pre-authorization completion transaction request to BASE24 ML Acquirer as a native-mode message, where it is received by the BASE24 XPNET and forwarded to the Device Handler with a message type of "M" or Merchant Link Financial transaction and message subtype "F" (Force Post).

- 13A-14-15 The Device Handler module receives the pre-authorization completion and first formats a SPDH response (Resp Code 000) to complete the transaction. It sends the response to the terminal to acknowledge that the messages was received.
Second, it formats an Advice transaction request 0220 into a POS Standard Internal Message (PSTM), with the DEST field set to "ML" to identify the transaction as belonging to an external FI and passes the transaction request as a 0220 message to the Router/Auth module.

The Router/Auth module logs the 0220 transaction as an approved transaction. It routes the completion to the Merchant Link Interface based on the routing parameters.

The PSTM reversal request is also appended with the original device reversal request (token CG).

- 16-17 The acquirer ML process receives the Advice PSTM (0220), retrieves the X0 token, creates a ML request and checks the connection with the Acceptor that needs to be UP and logically Logon ON.
If the above is true, it suspends the reversal transaction, sets an outbound timer and sends the reversal request (0220) (not including the PSTM) to the acceptor ML, prefixed by a message header in either SPDH or ISO format.

When the communications link with the Acceptor is down or the outbound timer expires, the Acquirer will write the completion to the Store And Forward file. Once the completion has been written, the "SAF CTR" flag on the Acquirer internal DPC table is incremented by one.

When the communication link is down between the Acquirer and the Acceptor, the systems continually check for the re-establishment of the link. Once the Acquirer ML interface receives a notification that the line is back up, the interface will first check the SAF file. If transactions reside in the SAF file, the Acquirer ML will forward these transactions first and decrease the SAF-CTR by one.

Every time a completion is placed on the SAF file no other incoming request from the same PINPAD id will be sent to the Acceptor until the SAF record has been removed otherwise, the transaction will be declined.

Note: The ML interface has a SAF cycle that is being driven by the MLCF file SAF timer and SAF retries parameter.

- 18-19 Upon receiving the transaction request, the ML Acceptor also removes the ML header and forward the SPDH pre-authorization completion transaction request to the Device Handler/Router/Auth as if the ML is a POS device.
The ML Acceptor handles real-time and SAF pre-authorization completions the same way.

If a PIN Pad mapping is required for this DPC, FID t (PIN Pad ID) is extracted from the request and mapped to a unique internal ID.

The Acceptor Device Handler, detecting the presence of FID t, uses the PIN Pad ID to retrieve a POS Terminal Data (PTD) record and processes the reversal transaction.

When the Device Handler/Router/Auth receives a pre-authorization completion it will determine if it is the issuer of the card. When the Acceptor is the issuer of the card, the Router/Auth will validate the hold, by comparing the Terminal Time of the completion with the Time of the pre-authorization hold.

The Router/Auth will also check the completion time without offsetting it to the terminal local time of the Acceptor. The request and completion times will be the same as the terminal location time.

The Router/Auth logs the completion to the PTLF file as 0220. The UAF/CAF hold and usage is updated. If the Acceptor approves or rejects the completion due security error, edit error, etc, the Device Handler will send a SPDH transaction response as acknowledgment to the Acquirer ML (with the appropriate response code depending on error condition) on the PSTM 0230 message.

The SPDH transaction response will not contain on-line terminal keys, since these keys will not be sent to the terminal, there is no need to include them in the response. The key usage and failure counters will be updated by the response to minimize the security exposure.

The Acceptor also sends a \$0 Issuer Reversal to the Authorize, Switch or Host in order to remove any "holds" which have been placed on the cardholder's account.

The Acceptor ML process retrieves the suspended transaction, deletes the inbound timer, creates a ML response based on the 0230 response message and sends it to the ML acquirer.

The Totals in the MTF file will be updated.

20

The ML Acquirer receives a transaction response (0230) for a real time or SAF cycle. Then, it deletes the outbound timer or the SAF record will be deleted.

The Totals in the MTF file will be updated.

The Acquirer ML interface will remove a completion from the SAF (a message type 0220) when one of the following conditions are met:

- A matching 230 is found
- The number of times that a 220 was sent out exceeds the limit set on the MLCF screen for that DPC, as specified by the “**MAX SAF RETRIES**” field.

If the (230) response contains a response code that is greater than "050" (transaction decline), the Acquirer ML process will not update the totals in the MTF file.

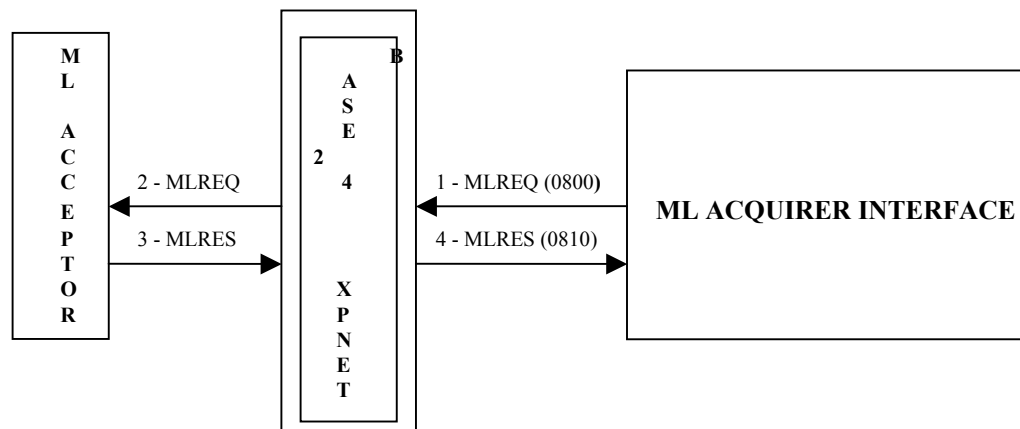
For both a Real-Time and a SAF pre-authorization completion, the Acquirer ML interface will drop the (230) response and will not send the response to the SPDH. The Acquirer SPDH had already sent the message response as acknowledgement to the terminal and logged the completion to the Acquirer PTLF file.

3.6. Network Management

In the diagram below, the transaction is a ML Network Management transaction. (See Section 2.4.2. The messages supported by the ML are:

- Logon
- Logoff
- Echo test
- Key exchanges

The network management messages can be initiated at either the Acceptor or the Acquirer DPC. The following flow is an example where the message is initiated by the Acquirer.



Steps

Processing

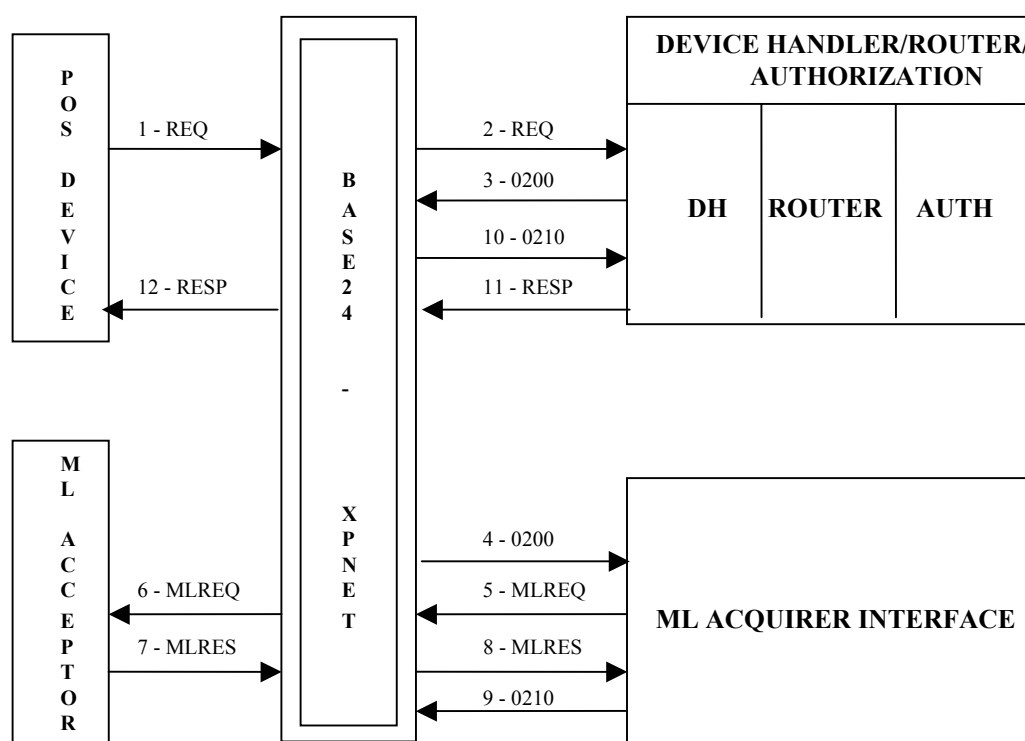
- 1-2 The ML Acquirer sends a network request to the ML Acceptor as an 0800 message.
- 3-4 According with the subtype of message request sent by the Acquirer the Acceptor will respond with a 0810 response message.

3.7. MACcing or PIN pad Resynchronization

In the diagram below, the device has been unable to get a transaction approved because of either a MACcing (898) or PIN pad (210) error.

The acceptor system keeps track of the number of failures and at the appropriate threshold, which is defined by Interac, will send a new PIN key or new MAC key. If a new MAC key is required then it will be sent in FID “H”, a new MAC key will be sent in FID “M”.

The diagram shows what happens when the threshold is exceeded and the **Acceptor** System decides that it is time to send a new key. As per Interac regulations, the Acceptor is in control of the encryption and PIN pad security.



Steps

Processing

- 1-2 The ML message is sent from the terminal with the incorrect MAC.
- 2-3 The Device Handler module processes the request as normal, ignoring the PIN Pad identifier (FID t) and formats the transaction request into a POS Standard Internal Message (PSTM), with the DEST field set to “ML” to identify the transaction as belonging to an external FI, adds the two ML tokens, sets a response timer, and passes the transaction request as a 0200 message to the Router/Auth module.

The Router/Auth module also processes the request as normal, retrieving the Retailer and card specific information and finding the ML BASE24 symbolic process name within the Auth Selection Table (AST) or Institution Definition File (IDF). The PSTM request is at this point fully formatted for authorization, and is also appended with the original device request (token CG).

- 4-5 The acquirer ML process receives the PSTM, retrieves the CG token, creates a ML request, suspends the transaction, sets an outbound timer, and sends the original request (not including the PSTM) to the acceptor ML, prefixed by a message header. This request may be in ISO format.
- 6-7 [At the acceptor site (In this diagram assume acceptor is another B24 system)].
- Upon receiving the transaction request, the acceptor ML removes the ML header and sets it's own inbound timer, suspends the transaction. If a PIN Pad mapping is required for this DPC, FID t (PIN Pad ID) is extracted from the request and mapped to a unique internal ID. The request, in the format of a normal transaction request, is sent to the DH as if the ML is a POS device.
- The Acceptor Device Handler, detecting the presence of FID t, uses the PIN Pad ID to retrieve a POS Terminal Data (PTD) record and processes the transaction in the current manner.
- Using the PTD record that was retrieved, the FIL determines that the threshold has now been exceeded. It generates the new working key and puts it into the appropriate FID. This message is then sent back to the ML acceptor interface with the error code in the SPDH header.
- The Acceptor ML process retrieves the suspended transaction, deletes the inbound timer, creates a ML response based on the 0210 response message and sends it to the ML acquirer.
- The Totals in the MTF file will not be updated, as this is an error.
- 8-9 The ML acquirer interface process retrieves the suspended record, reformats the response message into a PSTM 0210 response and CG token message, deletes the outbound timer, and sends the response message to the Device Handler/Router/Authorization process.
- 10 The Authorization module logs the transaction to the POS Transaction Log File (PTLF) including the CI token.

- 11-12 The Device Handler module reformats the 0210 response and CG token message into a SPDH native-mode format, deletes the response timer, and sends the response to the POS device.
- The POS device examines the incoming response, determines that a new working key has been sent. The terminal uses this key and then recalculates the encrypted value, formats a new message and sends it to the ML Acquirer.
- At this point normal processing resumes.

4. Appendix A - ISO 8583 to MLSPDH Mapping

BIT	LEN	TYP	FIELD DESCRIPTION	MAPPING
	2	a	Identifies message as belonging to the ML interface - has a constant value of "ML".	IDENTIFIER of ML Header
	4	n	Data Processing Centre number identifying the connection between the acquirer and the acceptor (value is the same for both parties).	DPC NUMBER of ML Header
	4	n	The message type has the following allowed values: 0100 = Authorization request 0110 = Authorization request response 0200 = Financial transaction request 0210 = Financial transaction request response 0220 = Financial transaction advice 0230 = Financial transaction advice response 0420 = Transaction reversal advice 0430 = Transaction reversal advice response 0500 = Reconciliation request 0510 = Reconciliation response 0600 = Administrative request 0610 = Administrative response 0800 = Network management, sign on 0810 = Network management, echo test Types are similar to those in the ML format, with the following exceptions: 0100/0110 corresponds to ML type 0200/0210 pre-auth transaction 0500/0510 corresponds to ML type 0800/0810 with nmm code 201 (see bit map #70) 0600/0610 administrative transactions correspond to ML type 0000	MESSAGE TYPE of ML Header
	16/8	an/b	The primary bit map indicates the presence or absence of data fields for elements 1 through 64. This 64 bit binary field may be translated to a 16 character hexadecimal string if indicated in the MLCF.	
1	16/8	an/b	The secondary bit map for elements 65 through 128. Format is the same as that of the Primary bit map (above).	

3	6	n	The ISO standard processing code, which consists of a two digit transaction code followed by to and from account types (unused account type is zero filled). The following on-line transaction types are supported: 00 Purchase 02 Purchase Adjustment 03 Pre-auth 09 Purchase/Cash Back 17 Pre-auth Completion 20 Merch. Return 22 Merch. Return Adjustment 30 Balance Inquiry The following account types are supported: 10 Savings 20 Chequing 30 Credit On administrative transactions (type 0600/0610), this field will contain the BASE24 defined transaction code, left justified, zero filled.	Transaction code of SPDH Header FID D - Appl Acct Type
4	12	n	Amount of the transaction	FID B - Amount 1
7	10	n	Date and time (local) at which the transaction is transmitted to the authorizing entity. (Format: MMDDHHMMSS)	DATE and TIME of ML Header.
11	6	n	System trace number associated with this transaction.	TRACE NUMBER of ML Header
12	6	n	Local terminal transaction time when the transaction originated. (Format: HHMMSS)	Local transaction TIME of the SPDH Header
13	4	n	Local terminal transaction date when the transaction originated. (Format MMDD)	Local transaction DATE of the SPDH Header
15	4	n	Date on which the transaction will be settled from the point of view of the merchant. (Format: MMDD). On a reconciliation transaction, this will indicate the date for which totals are requested.	FID K - Business Date Posting Date of NMM ML message.
17	4	n	BASE24 posting date. (Format: MMDD)	FID u - Posting Date
35	2 37	n an	Entered TRACK2, not including the start and end sentinels contained on the magnetic stripe. The start sentinel, which indicates swiped or manual entry at the POS device level, will be set based on the POS condition code (Bit Map 60).	FID q - Track2 / Customer
37	12	n	Retrieval reference number. This corresponds to the 10 digit device sequence number (9 digit # followed by a reset flag) as right justified, zero filled.	FID h - Sequence Number
38	6	an	Authorizing host's transaction identification.	FID F - Approval Code

39	2	n	ISO response code associated with this transaction.	FID X - ISO Response Code RESPONSE CODE of MNL NMM message
41	8	an	Terminal identification.	FID t - Acceptor PIN Pad ID
42	15	an	Card acceptor identifier, corresponding to the 12 byte retailer ID supported by the POS device (left justified, blank filled).	FID d - Retailer ID
43	40	an	On transaction requests, this field will contain the date of birth (YYMMDD) of the cardholder. On replies, this will contain the location of the POS PIN Pad (25 bytes) according to the acceptor. In each case, this fixed field will be left justified and blank filled.	FID k - Terminal Location/DOB
44	3 1	n n	Contains the language code (one byte numeric) to be used for processing the current transaction. The length of this field will always be 1.	FID U - Language Code
48	3 48	n an	This private field carries the SPDH standard message header fields. It will always be 48 characters long.	SPDH Header Fields: dev-addr xmit-num term-id clerk-id date time req-type req-subtype tran-code proc-flags resp-code
52	16	an	Personal Identification Number.	FID b - PIN/Customer
53	2 50	n an	Security related control information consists of a one character MAC key index, followed by: On on-line responses, any new working keys are included as an ISO subfield, consisting of a one byte binary (or hex - see bit #1) bit map indicating which keys are included, followed by the new key values (in order of FID). On ML key exchange, the working key check digits, followed by the working key itself.	MAC INDEX of ML Header FID H - KMAC FID I - KME FID M - KPE KMAC of NMM Key Exchange and CHECK DIGIT.
54	2 48	n an	Additional data - on requests, this field may contain FID j. On responses, FID g.	FID j - State Code FID g - Response Display
57	3 100	n an	Product SFIDs. E-commerce flag. It is optional on request contain FID 6 SFID F.	FID 6 – Product SFIDs SFID F – e-comm flg
58	3 100	n an	Address verification related information. It is optional. On request contain FIDs A and Y. On response FIDs A, Y and Z	FID A – Billing address FID Y – Postal code FID Z – Status code

60	3 7	n n	The ISO POS condition code (length 7) applying to this transaction, which will be built from an ML/SPDH message according to the following algorithm: <table><tr><th>BYTE</th><th>VALUE</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>3</td></tr><tr><td>3</td><td>0 if FID e = 01 or 08 1 otherwise</td></tr><tr><td>4</td><td>0 if FID q begins with "M" 1 otherwise</td></tr><tr><td>5</td><td>1 if FID e = 02 0 otherwise</td></tr><tr><td>6</td><td>3</td></tr><tr><td>7</td><td>1 if FID e = 03,09, or 11 2 if FID e = 10 0 otherwise</td></tr></table> In the other direction, FID e will be created based on the following values, with a default of 00 (values lower in the table supersede those above): <table><tr><th>VALUE</th><th>FID e</th></tr><tr><td>BYTE(4) = 0</td><td>05</td></tr><tr><td>BYTE(2) = 0</td><td>01</td></tr><tr><td>BYTE(5) = 1</td><td>09</td></tr><tr><td>BYTE(5) = 2</td><td>10</td></tr></table>	BYTE	VALUE	1	1	2	3	3	0 if FID e = 01 or 08 1 otherwise	4	0 if FID q begins with "M" 1 otherwise	5	1 if FID e = 02 0 otherwise	6	3	7	1 if FID e = 03,09, or 11 2 if FID e = 10 0 otherwise	VALUE	FID e	BYTE(4) = 0	05	BYTE(2) = 0	01	BYTE(5) = 1	09	BYTE(5) = 2	10	FID e - POS Condition Code FID q - Track 2
BYTE	VALUE																													
1	1																													
2	3																													
3	0 if FID e = 01 or 08 1 otherwise																													
4	0 if FID q begins with "M" 1 otherwise																													
5	1 if FID e = 02 0 otherwise																													
6	3																													
7	1 if FID e = 03,09, or 11 2 if FID e = 10 0 otherwise																													
VALUE	FID e																													
BYTE(4) = 0	05																													
BYTE(2) = 0	01																													
BYTE(5) = 1	09																													
BYTE(5) = 2	10																													
61	3 28	n an	Transaction ML reconciliation control fields (28 bytes), left justified, blank filled.	Reconciliation Entity, Settlement Identifier, Card Type of ML Header.																										
62	3 54	n an	An ISO subfield, consisting of a one byte binary (or hex - see bit # 1) bit map indicating which of the two fields are included, followed by the new data (in order of FID). This field contains the entered administrative PIN (blank if not present) and administrative card TRACK2 (excluding start and end sentinels - will be assumed to be swiped).	FID c - PIN / Supervisor FID r - Track 2 / Supervisor																										
63	3 1	n an	One character field indicating whether the card swiped at the POS device is a debit ("D") or a credit ("C") card	FID R - Card Type																										
64	16	an	Message authentication code. This field will contain (left justified, blank filled) the transaction MAC generated by the POS device's PIN Pad.	FID G -Authentication Code																										
70	3	n	Network management information code identifying the status of the network. Allowed values are identical to those in the ML Header, with the exception of code "201" (reconciliation). 0800 messages with that code correspond to ISO 0500 messages (see message type).	NMM CODE of ML Network Management Message																										
74	10	n	Total number of credit transactions processed.	CREDIT COUNT of ML totals.																										

75	10	n	Total number of adjustment transactions processed.	ADJUSTMENT COUNT of ML totals.
76	10	n	Total number of debit transactions processed.	DEBIT COUNT of ML totals.
86	16	n	Total amount of all credit transactions processed.	CREDIT AMOUNT of ML totals.
87	16	n	Total amount of all adjustment transactions processed.	ADJUSTMENT AMOUNT of ML totals.
88	16	n	Total amount of all debit transactions processed.	DEBIT AMOUNT of ML totals.
91	1	n	File update code directing the maintaining system on how to process the record.	FID P - Draft Capture Flag
95	42	n	Original or replacement transaction amount. BASE24 currently uses only the first 12 bytes.	FID C - Amount 2
102	2 28	n n	Account number associated with the "FROM" side of a transaction.	FID E - Appl Acct Num
103	2 28	n n	Account number associated with the "TO" side of a transaction.	FID E - Appl Acct Num
104	3 24	n an	A verbal description of the transaction, if transmitted, for billing purposes.	FID s - Transaction Description
120	3 18	n n	Available balance.	FID J - Available Balance
121	3 27	n an	Cheque guarantee and verification fields. These are grouped as common fields. BASE24 does not currently support this functionality.	FID L - Check Type / Category FID O - Customer ID Type FID N - Customer ID
122	3 455	n an	On administrative requests, mail and download fields are used together (i.e. FID W is not included without FID V) and so are grouped together. FID V (fixed length) only will be included on requests. FID W is a variable length value that will be appended on partial or full download response messages. On financial requests, this bit map contains FID f.	FID V - Mail / DDL Key FID W - Mail / DDL Text FID f - PIN length or receipt data
123	3 20	n an	Invoice information. Field length will indicated the presence of FID T. It is assumed that FID T will not appear if FID S is not present.	FID S - Invoice Number FID T - Original Invoice Number
124	3 12	n n	The original sequence number is sent by the terminal on transactions for which the sequence number was reset. To preserve field formats (see Bit Map 37), this corresponds to the 9 digit device sequence number in the following format: 00<device # - 9 bytes>0	FID i - Original Sequence Number

125	3 324	n an	Terminal totals will be formatted as an ISO subfield, consisting of a one byte binary (or hex - see bit #1) bit map indicating which totals are included in this field. This will be followed by one or more occurrences of the indicated totals, ordered alphabetically by FID.	FID l - Totals/Batch FID m - Totals/Day FID n - Totals/Clerk FID o - Totals/Shift
126	3 16	n an	An optional field containing data that will be reflected back to the device on responses. No changes will be made to this field.	FID Q- Echo Data
127	3 80	n an	User defined optional data sent by the POS device.	FID a- Optional Data
128	16	an	Secondary message authentication code, if present, contains the ML message MAC (if ML MACcing is being used), which will be built from the following fields: ML message indicator DPC number Message type (translated) Bit Map 11 (trace number) Bit Map 7 (transmission date/time) Bit Map 43 (ML reconciliation data - blanks if not included)	MAC of ML Header.

[illegible]

4	12	N	<u>AMOUNT 1</u> Primary amount of the transaction, right justified, zero filled. This field is mapped tot FID B which supports a larger field size. Shell uses a maximum field length of eight so the field must be filled with leading zeroes. The MAC calculation will use the complete field to derive the value.	FID B
7	10	N	<u>TRANSMISSION DATE/TIME</u> It is the time (local) at which the transaction is transmitted to the authorizing entity. The first four characters of this field correspond to the month and day fields (MMDD) of the Merchant Link Header date. This is expressed, as the GMT, in the following format: Transmission Date (MMDD) PIC 9(4) Transmission Time (HHMMSS) PIC 9(6) In all messages received from BMO this value will be used to determine the correct switch posting date. Should the transmission date be one greater than the switch posting date, the posting date changes to the transmission date. If the time is greater than the switch cutover time, the posting date move to tomorrow's date.	ML switch generated
11	6	N	<u>TRACE NUMBER</u> It is a direct field mapping to/from the Merchant Link Header trace number.	ML switch generated
12	6	N	<u>TRANSACTION TIME</u> Mapped from the SPDH header transaction time.	SPDH Trans. Time
13	4	N	<u>TRANSACTION DATE</u> The month and day (MMDD) obtained from the SPDH header transaction date.	SPDH Trans. date
14	4	N	<u>EXPIRATION DATE</u> The year and month (YYMM) this extracted from the TRACK2. This corresponds to the first four characters after the equal sign (=) in FID q. If the Track 2 field is present this field is not required.	FID q
18	3	N	<u>MERCHANT TYPE</u> It is not currently required	

22	3	N	<u>POS ENTRY MODE</u> It is field is derived from the first two characters beginning at the start sentinel of the TRACK2. The start sentinel, which indicates swiped ("S") or manual ("M") entry at the POS device level. If the card was swiped a value of "02" is entered. A manually entered card has a value of "01". The third character is currently set to zero (0) on credit transactions. On debit transactions this field is set to a "1" (PIN capable).	FID q
35	2 37	N AN	<u>TRACK2</u> It is the contents of cards magnetic stripe not including the start and end sentinels. This corresponds to FID q without its start, end sentinel and LRC. The start sentinel is indicated by: "S" swiped or "M" manual entry at the POS device and the end sentinel is always "?". The first two bytes corresponds to the value entered in Bit Map 22 POS Entry Mode.	FID q
37	12	N	<u>RETRIEVAL REFERENCE NUMBER</u> Corresponds to the Invoice Number FID in the SPDH message. The value is defined as CCCCCTnnnn where: C - Terminal location number T - Terminal number n - Sequential number This value was used specifically to meet Shell's requirements	FID S
38	6	AN	<u>APPROVAL CODE</u> It is authorizing host's transaction identification which corresponds to the eight character BASE24 approval code. When FID F is mapped to this Bit Map, it's left justified and the last two characters are truncated. When FID F is created from the Bit Map, it's left justified and blank filled.	FID F
39	2	N	<u>ISO RESPONSE CODE</u> It is the ISO response code associated with this transaction. For financial messages this field corresponds to FID X and for network management or reconciliation messages the value is the response code. The next section contents a table which maps ISO response code to BASE24 ones. This section also clarifies when these are to be expected.	FID X /ML response code
41	8	AN	<u>CARD ACCEPTOR TERMINAL ID</u> It is the acceptor's PIN Pad, corresponding to FID t found in the POS device messages. Due to ISO format limitations, this field length is limited of eight characters but FID t contains up to 16 characters.	FID t

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54	2 40	N AN	<u>ADDITIONAL DATA</u> It is not currently required.	
60	3	N	<u>ADVICE REASON CODE</u> It is the reason the transaction Advice(reversal) was generated. This value will correspond to BMO ML requirements. The following value will be used: 01 - Time out. Used when the transaction timer expires at either the POS device, the transaction router or the BMO ML. Also used when the terminal disconnects. The transaction will be returned from the terminal with a value of "C" in the transaction type of the SPDH header. 02 - Destination not available. Used when the PIN PAD or Printer are not responding at the device. The transaction will be returned from the terminal with a value of "P" in the transaction type of the SPDH header. 03 - Fraudulent Transaction. Not used 30 - MAC failure. Used when the PIN PAD MAC verify procedure is failed on a transaction. The transaction will be returned from the terminal with a value of "R" in the transaction type of the SPDH header.	ML switch generated
70	3	N	<u>NETWORK MANAGEMENT CODE</u> It is only included for network management messages (type 0800/0810), this is the ML network management code. Possible value are: 001 - Sign-on DPC 002 - Sign-off DPC 301 - Echo Test	ML switch generated
73	6	N	<u>DATE OF BIRTH</u> It is not currently required.	

90	20	N	<u>ORIGINAL DATA ELEMENTS</u> are present for reversals (0420) only. This field retrieves the following fields from the original BMO ISO request: Org Msg Type ID (ISO header) PIC 9(4) Org System Trace Audit Num (BM-11) PIC 9(6) Org Local Transaction Time (BM-12) PIC 9(6) Org Local Transaction Date (BM-13) PIC 9(4)	the Original ISO message
91	1	N	<u>FILE UPDATE CODE</u> It is not currently required	
94	2	AN	<u>FILE UPDATE CODE</u> It is not currently required	
95	12	N	<u>REPLACEMENT AMOUNTS</u> It is the original or replacement transaction amounts, of which only the first twelve digits are used, mapped to/from FID C (right justified, zero filled). Although this field is not currently used it is required for pre-authorization purchase completions.	FID C
96	8	AN	<u>MESSAGE SECURITY CODE</u> It is not currently required.	
101	2 17	N AN	<u>FILE NAME</u> It is not currently required.	
102	2 22	N AN	<u>CUSTOMER ID</u> It is not currently required.	
104	8	N	<u>OPERATOR ID</u> It is the first eight bytes of the Terminal ID from the SPDH header (PTDF). This field is padded with blanks and left justified	TERM ID of the SPDH header
120	3 999	N AN	<u>RECORD DATA</u> It is not currently required.	
122	3 999	N AN	<u>MERCHANT SETTLEMENT AMOUNTS</u> It is not currently required.	

126	2 99	N AN	<u>MERCHANT ORIGINAL ELEMENTS</u> It is not currently required.	
127	2 99	N AN	<u>PRIVATE DATA</u> It is not currently required.	
128	8	AN	<u>MAC CODE</u> It is used in financial requests and responses only. This value is not required for Network Management or Administrative transactions nor is it required for financial reversals or credit transactions. The message authentication code is built from the following fields, if present, in the following order: Processing Code (Bit Map 3) PIC X(6) Transaction Amount (Bit Map 4) PIC 9(12) Track II Data (Bit Map 35) PIC X(39) Variable length Retrieval Reference Number (Bit Map 37) PIC X(12) Response Code (Bit Map 39; Response only) PIC X(2) Security Related Data (Bit Map 53) PIC X(51) Variable length Replacement Amount (Bit Map 95) PIC X(12)	FID G